

documentation of ezFIO, a user-friendly graphical interface or wrapper designed to simplify the usage of FIO.

He didn't stop at benchmarking tools. The world of open source had opened a new gateway. "In 2015, there was a requirement to create a test bed where I had to set up more than 50 VMs and configure them in a certain way. I came across this tool, Ansible, which allowed me to automate and update software across all the machines. I learned that within 10-12 days! I had to share my feat with the world," he remarks.

With experience in documentation and significant contributions to the tools he had been using, Suman took his findings to the PyCon conference in Delhi and delivered a presentation on infrastructure automation for testing using Ansible and Vagrant. "Before that, whenever I attended a conference, it was always as a representative of a company. This was the first time I was representing myself and my work," he recalls proudly.

The conference connected Suman with many people from open source domains beyond storage and systems. "I was in Bengaluru engaging with various user groups like the Python community, AWS User Group, and the Docker community. I got to work on numerous open source projects, gaining in-depth learning, with source code and documentation readily available. Gradually, I began participating in more conferences, such as that of the Storage Networking Industry Association (SNIA)," he reveals.

The appeal of open source did not stop at community engagement. It was the learning. "Open source allows you to learn whatever you want, even if it's not your core area. You ask questions, and engineers from all over the world can answer them. If you use anything proprietary, you're often bound by SLAs, and you may not even be eligible to raise a request without an enterprise licence; with open source, I have the whole world who can contribute to my idea or product," he states.

His involvement with open source led him to join AWS as a principal developer advocate for India to create and share content with the community. "When I joined AWS, there were roughly 12 to 15 user groups. In two to three years, we scaled that number to more than 25," adds Suman.

Speaking about his community engagement, Suman reveals that since he joined AWS, the instances of speaking at tech conferences have grown manifold. Be it the Open Data Science Conference, PyCon, or even the Python Software Foundation, Suman has been completely immersed in educating the community about his work in open source, which lately revolves around Python, cloud, and machine learning.

"Today, I'm working on fine-tuning large language models (LLMs) and retrieval-augmented generation (RAG). We use many open source tools, such as LlamaIndex, LangChain, and pgvector (an extension of PostgreSQL). At this point, my whole work revolves around open source tools and other publicly available models," he reveals.

Advocacy and engagement

With his work around machine learning, Suman advocates the importance of responsible AI. "We need to understand that while working with machine learning, especially with any tool in the generative AI space, we

should be responsible and not exploit open source or mix it with anything that has licensing complexities. For this, good education and awareness are crucial," he advises.

The challenge does not stop at responsible and ethical use of AI in open source. Having worked with the open source community across the globe, Suman believes that sustainability is one of the biggest challenges. "We need to ensure that what we build solves a critical problem. If it is a problem that has already been solved but in a different way, it may not be sustainable, even if you get funding," he shares.

Educating and raising awareness about open source on a global scale is Suman's enduring commitment. "Forums such as NumFocus and the Python Software Foundation contribute to major open source projects around data science and Python in general. Funding and sponsorship for such projects should come from academia, industry, and individuals, in whatever capacity," he stresses.

The community leader highlights the need to mentor and educate engineers about open source, right from the start. "Had I known about open source and community contributions in college, I would be a different person today. I learned it during my mid-journey, but the earlier you start, the less you have to struggle as you grow," he shares.



At the AWS Gen AI Loft in San Francisco in 2024